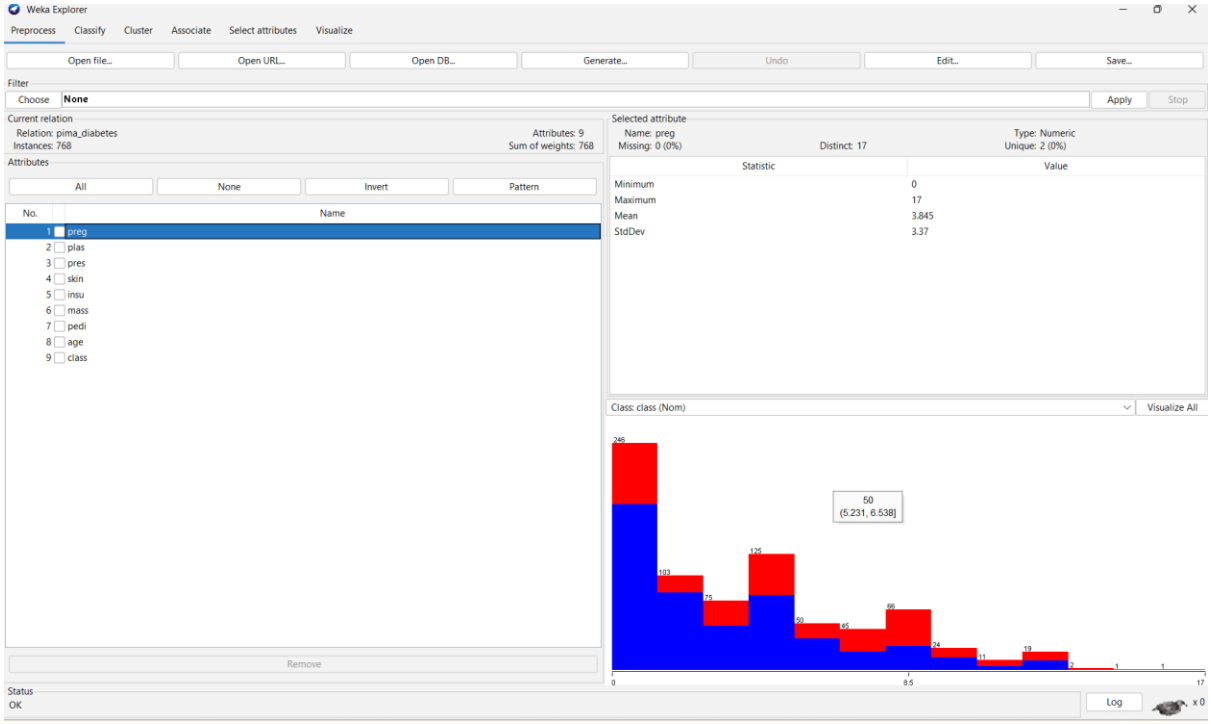
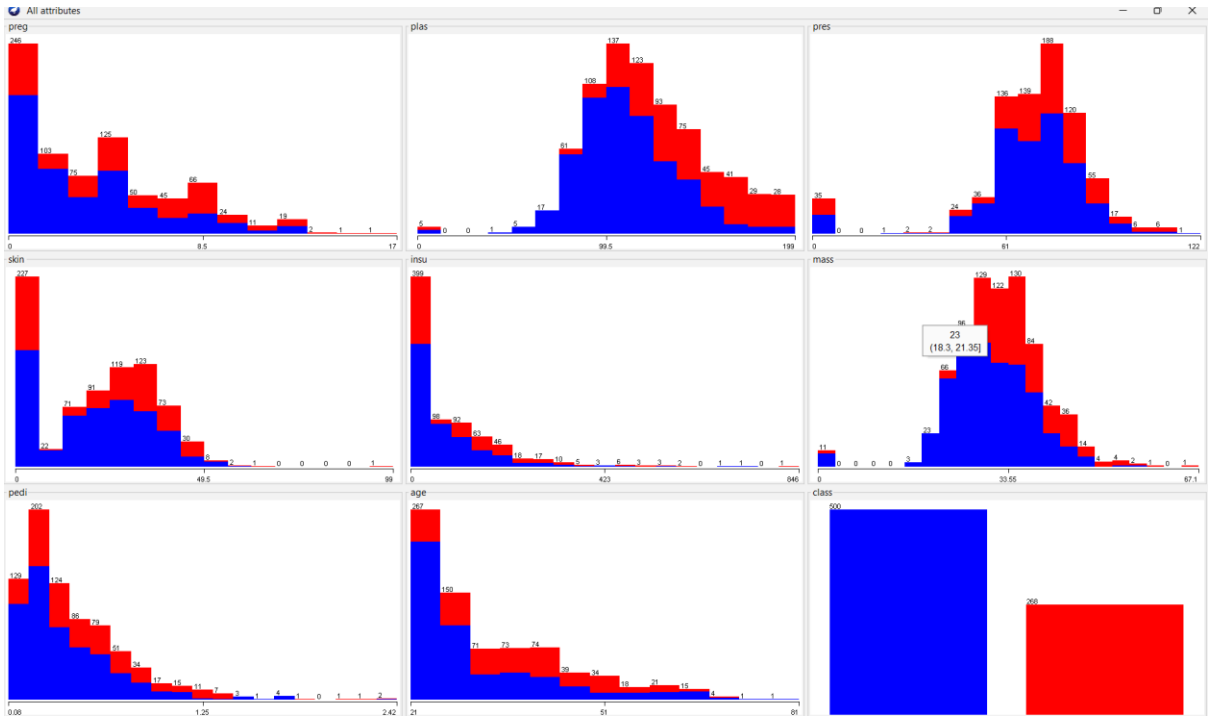
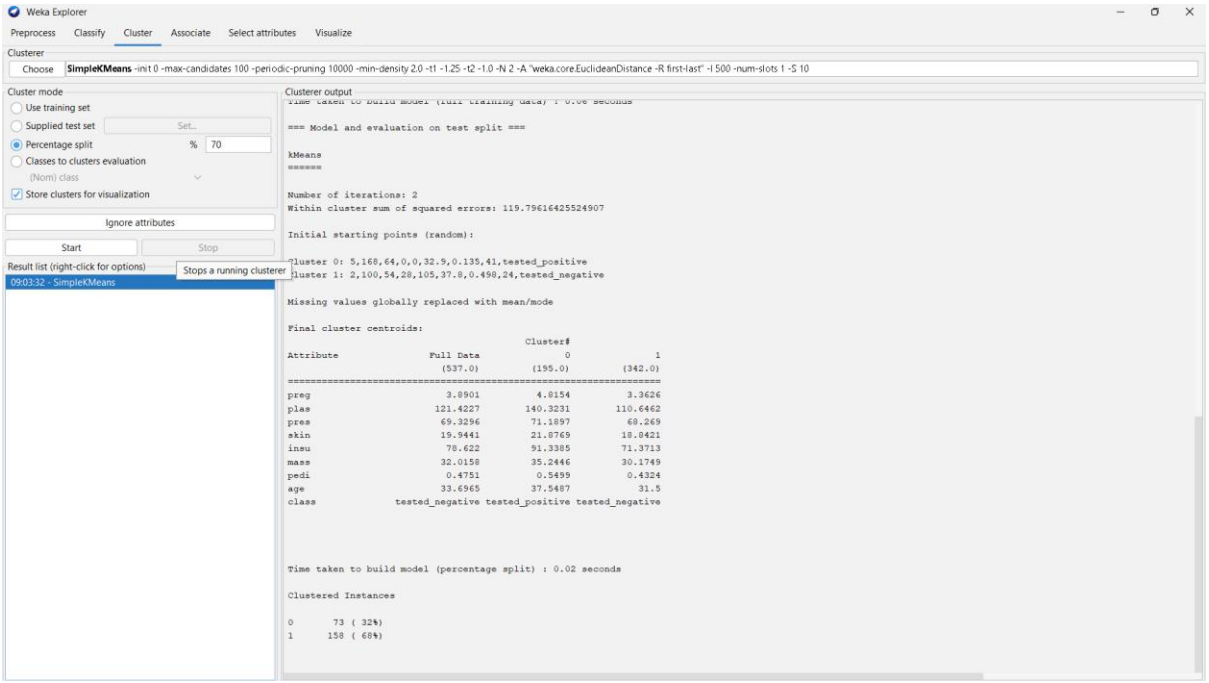
EXPERIMENTS OF 21 TO 30

21.



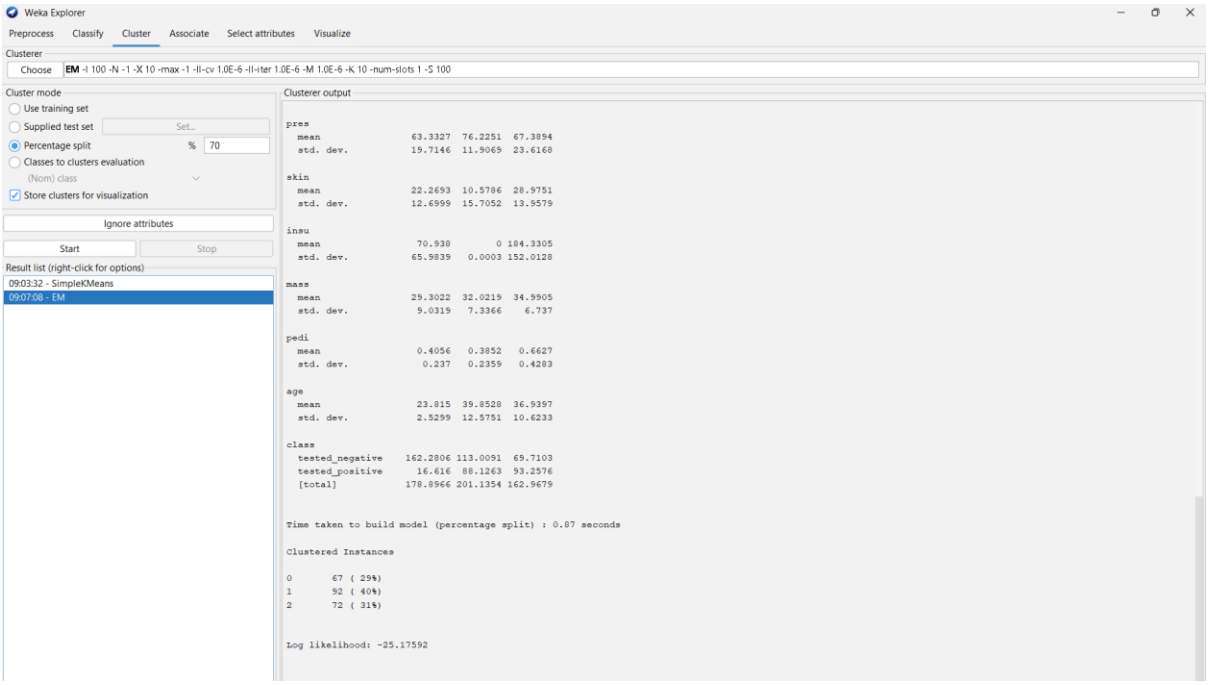


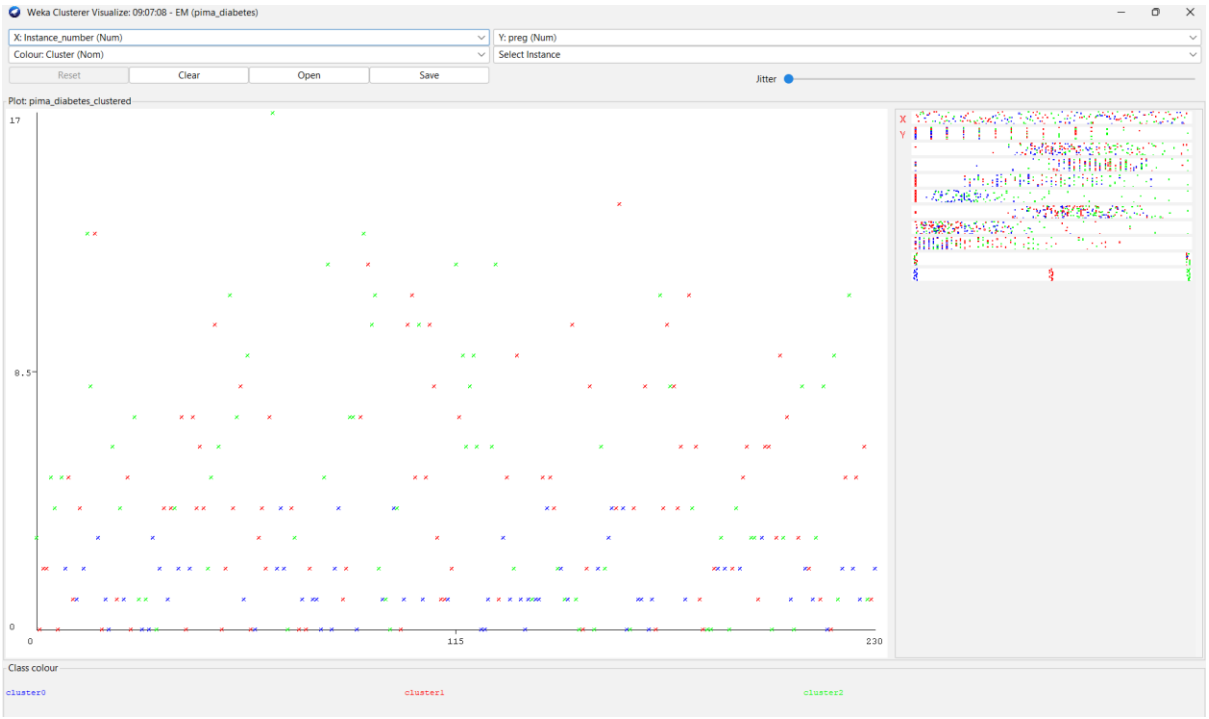
22.


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
%
70
Classes to clusters evaluation
(Nom) class
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
09:03:32 - SimpleKMeans
Stops a running clusterer
Clusterer output
=== Model and evaluation on test split ===
kMeans
======
Number of iterations: 2
Within cluster sum of squared errors: 119.79616425524907
Initial starting points (random):
Cluster 0: 5,168,64,0,0,32.9,0.135,41,tested_positive
Cluster 1: 2,100,54,28,105,37.8,0.498,24,tested_negative
Missing values globally replaced with mean/mode
Final cluster centroids:
Cluster#
Attribute Full Data 0 1
(537.0) (195.0) (342.0)
preg 3.8901 4.8154 3.3626
plas 121.4227 140.3231 110.6462
pres 69.3296 71.1897 68.269
skin 19.9441 21.8769 18.8421
insu 78.622 91.3385 71.3713
mass 32.0158 35.2446 30.1749
pedi 0.4751 0.5499 0.4324
age 33.6965 37.5487 31.5
class tested_negative tested_positive tested_negative
Time taken to build model (percentage split) : 0.02 seconds
Clustered Instances
0 73 (32%)
1 158 (68%)

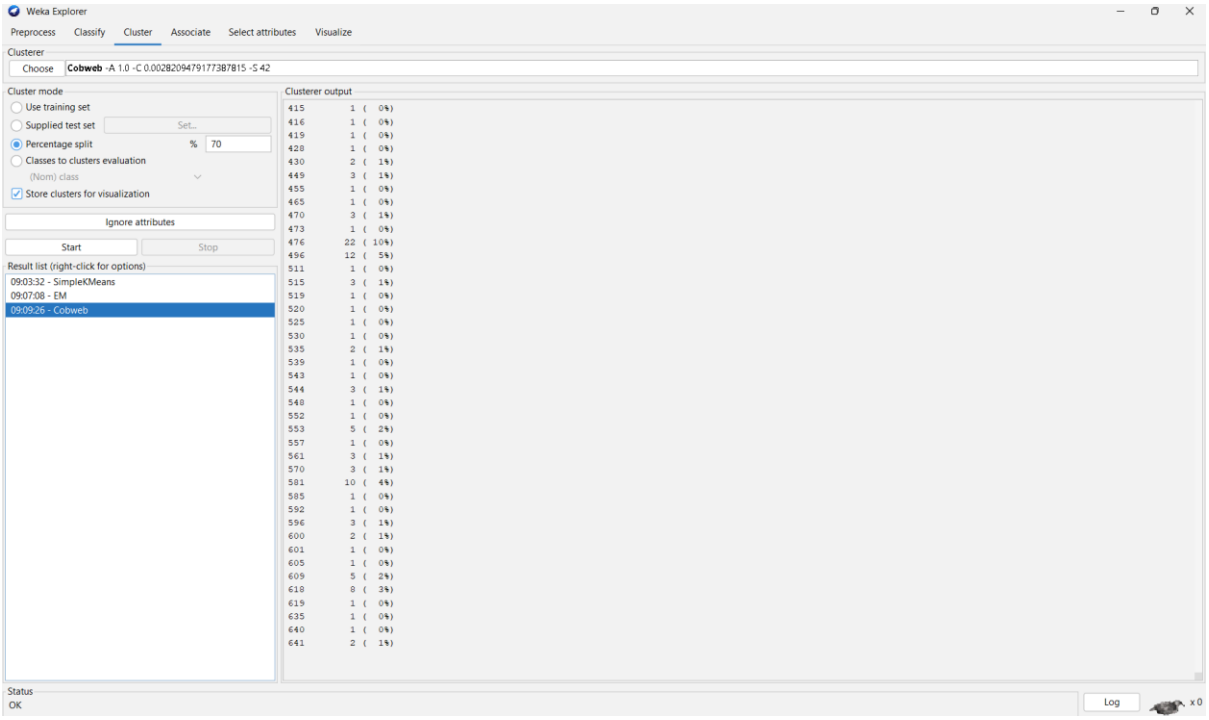

Weka Clusterer Visualize: 09:03:32 - SimpleKMeans (pima_diabetes)
X: Instance_number (Num)
Y: preg (Num)
Colour: Cluster (Nom)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: pima_diabetes_clustered
17
8.5
0
0
115
230
Class colour
cluster0
cluster1

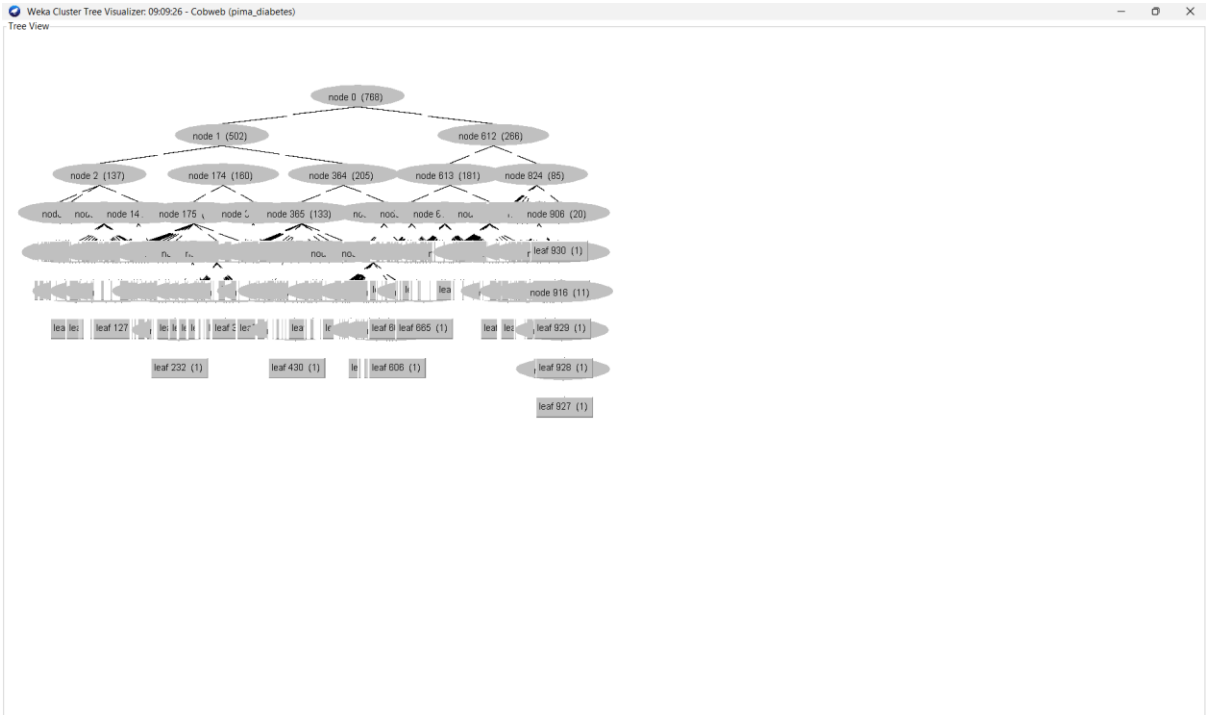
23.

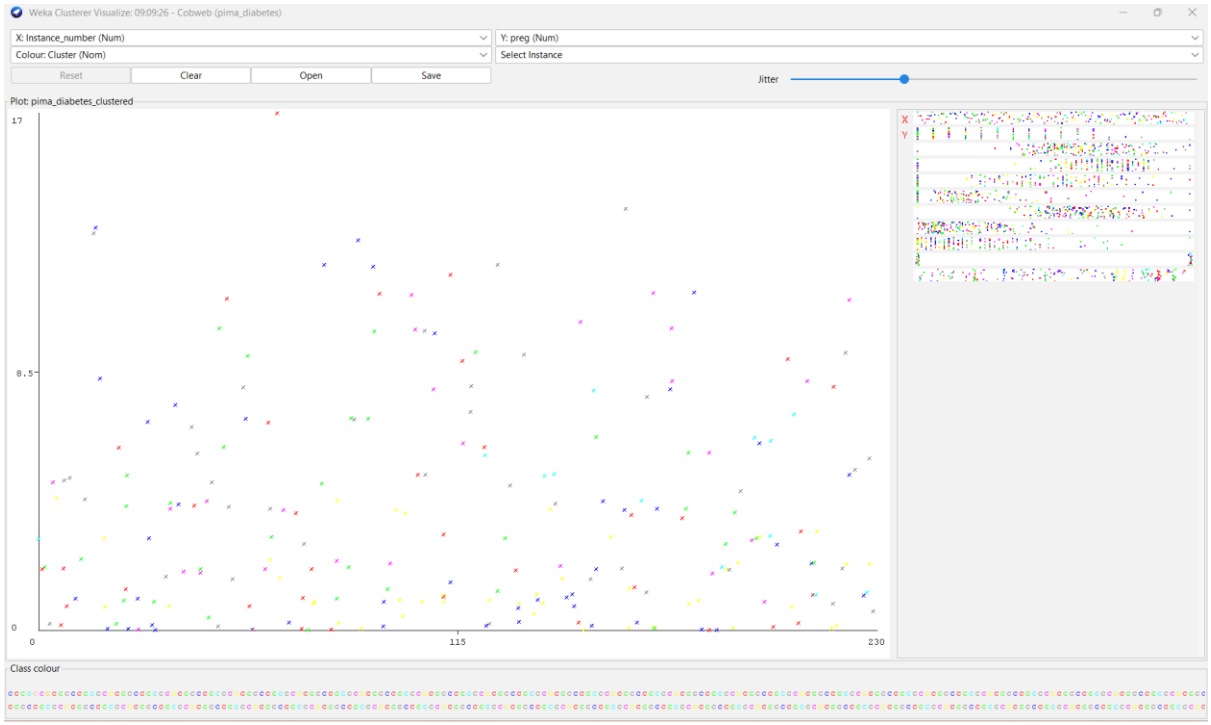

Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
EM -I 100 -N -1 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 100
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
%
70
Classes to clusters evaluation
(Nom) class
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
09:03:32 - SimpleKMeans
09:07:08 - EM
Clusterer output
pres
mean 63.3327 76.2251 67.3894
std. dev. 19.7146 11.9069 23.6168
skin
mean 22.2693 10.5786 28.9751
std. dev. 12.6999 15.7052 13.9579
insu
mean 70.930 0 184.3305
std. dev. 65.9839 0.0003 152.0128
mass
mean 29.3022 32.0219 34.9905
std. dev. 9.0319 7.3366 6.737
pedi
mean 0.4056 0.3852 0.6627
std. dev. 0.237 0.2359 0.4283
age
mean 23.815 39.8528 36.9397
std. dev. 2.5299 12.5751 10.6233
class
tested_negative 162.2806 113.0091 69.7103
tested_positive 16.616 88.1263 93.2576
[total] 178.8966 201.1354 162.9679
Time taken to build model (percentage split) : 0.07 seconds
Clustered Instances
0 67 (29%)
1 92 (40%)
2 72 (31%)
Log likelihood: -25.17592


Weka Clusterer Visualize: 09:07:08 - EM (pima_diabetes)
X: Instance_number (Num)
Y: preg (Num)
Colour: Cluster (Nom)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: pima_diabetes_clustered
17
8.5
0
0
115
230
Class colour
cluster0
cluster1
cluster2

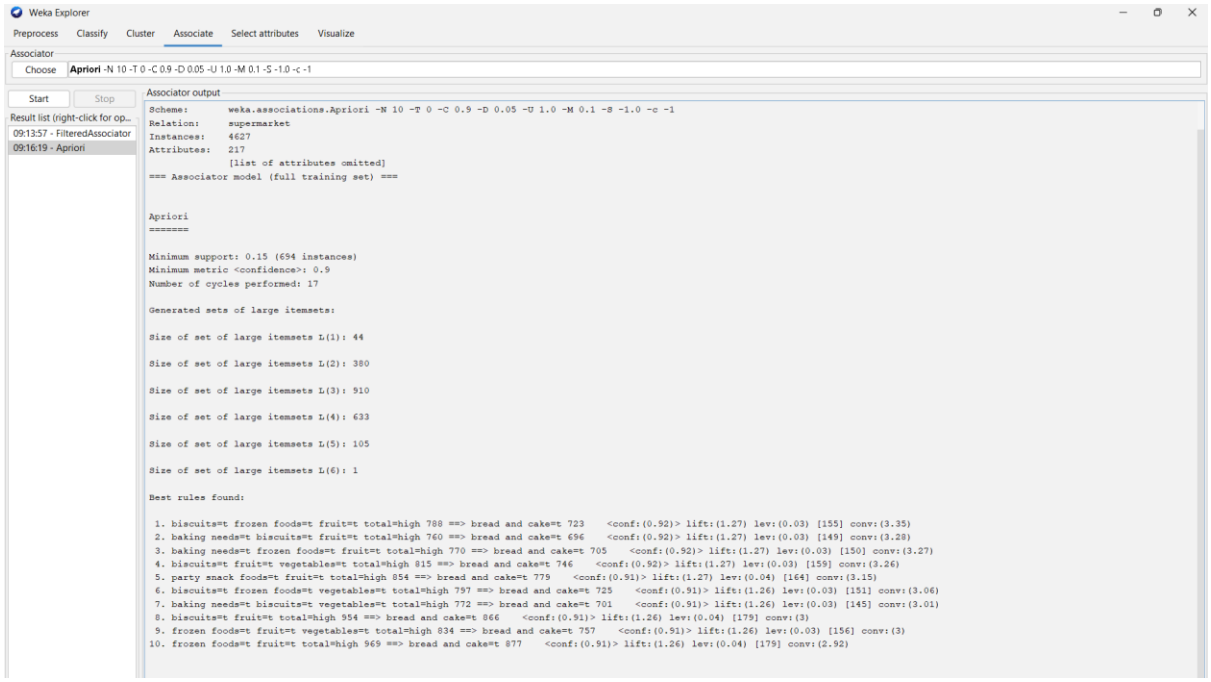
24.


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
Cobweb -A 1.0 -C 0.0028209479177387815 -S 42
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
%
70
Classes to clusters evaluation
(Nom) class
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
09:03:32 - SimpleKMeans
09:07:08 - EM
09:09:26 - Cobweb
Clusterer output
415 1 (0%)
416 1 (0%)
419 1 (0%)
428 1 (0%)
430 2 (1%)
449 3 (1%)
455 1 (0%)
465 1 (0%)
470 3 (1%)
473 1 (0%)
476 22 (10%)
496 12 (5%)
511 1 (0%)
515 3 (1%)
519 1 (0%)
520 1 (0%)
525 1 (0%)
530 1 (0%)
535 2 (1%)
539 1 (0%)
543 1 (0%)
544 3 (1%)
548 1 (0%)
552 1 (0%)
553 5 (2%)
557 1 (0%)
561 3 (1%)
570 3 (1%)
581 10 (4%)
585 1 (0%)
592 1 (0%)
596 3 (1%)
600 2 (1%)
601 1 (0%)
605 1 (0%)
609 5 (2%)
610 8 (3%)
619 1 (0%)
635 1 (0%)
640 1 (0%)
641 2 (1%)
Status
OK
Log
x 0

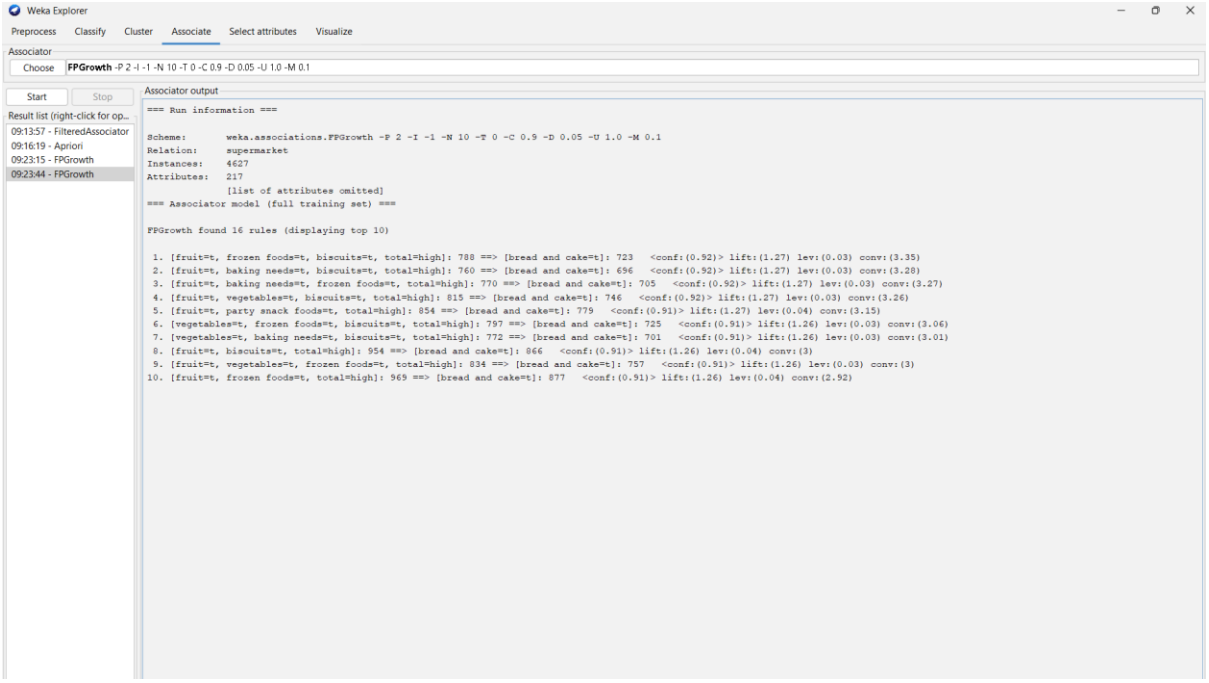

Weka Cluster Tree Visualizer: 09:09:26 - Cobweb (pima_diabetes)
Tree View
node 0 (768)
node 1 (502)
node 612 (266)
node 2 (137)
node 174 (160)
node 364 (205)
node 613 (181)
node 824 (85)
node 365 (133)
node 906 (20)
leaf 930 (1)
node 916 (11)
leaf 127
leaf 665 (1)
leaf 929 (1)
leaf 232 (1)
leaf 430 (1)
leaf 606 (1)
leaf 928 (1)
leaf 927 (1)


Weka Clusterer Visualize: 09:09:26 - Cobweb (pima_diabetes)
X: Instance_number (Num)
Y: preg (Num)
Colour: Cluster (Nom)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: pima_diabetes_clustered
17
8.5
0
0
115
230
Class colour

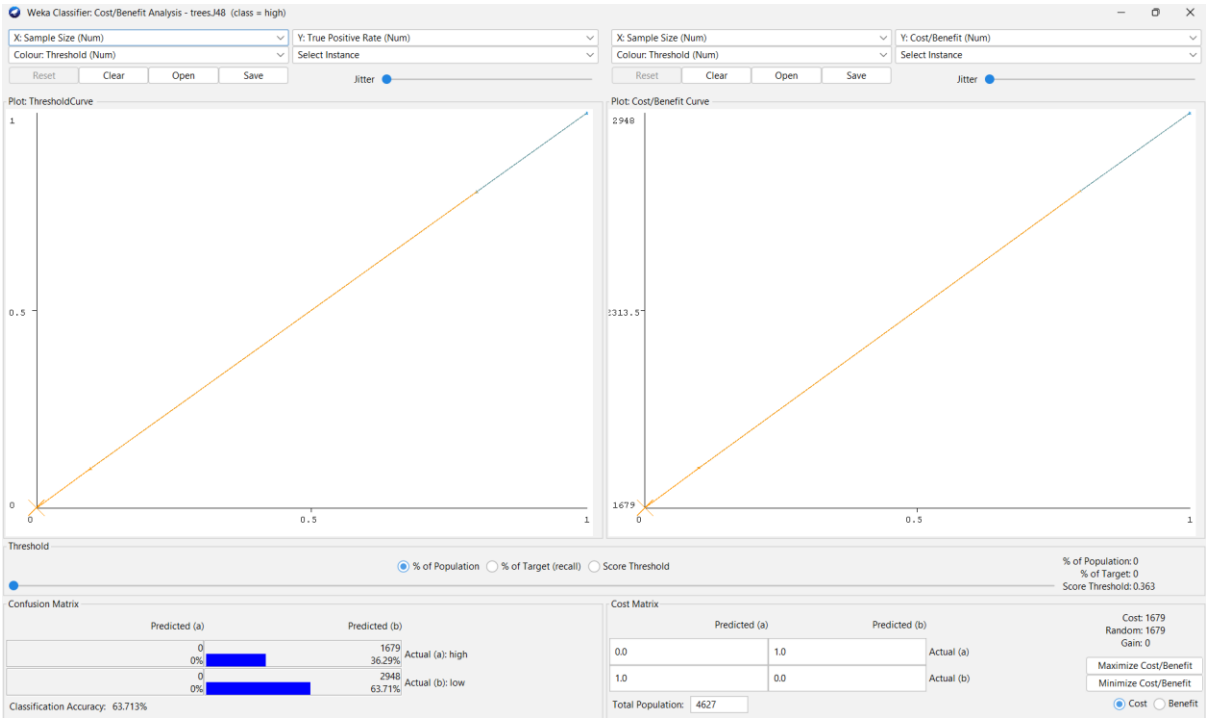
25.

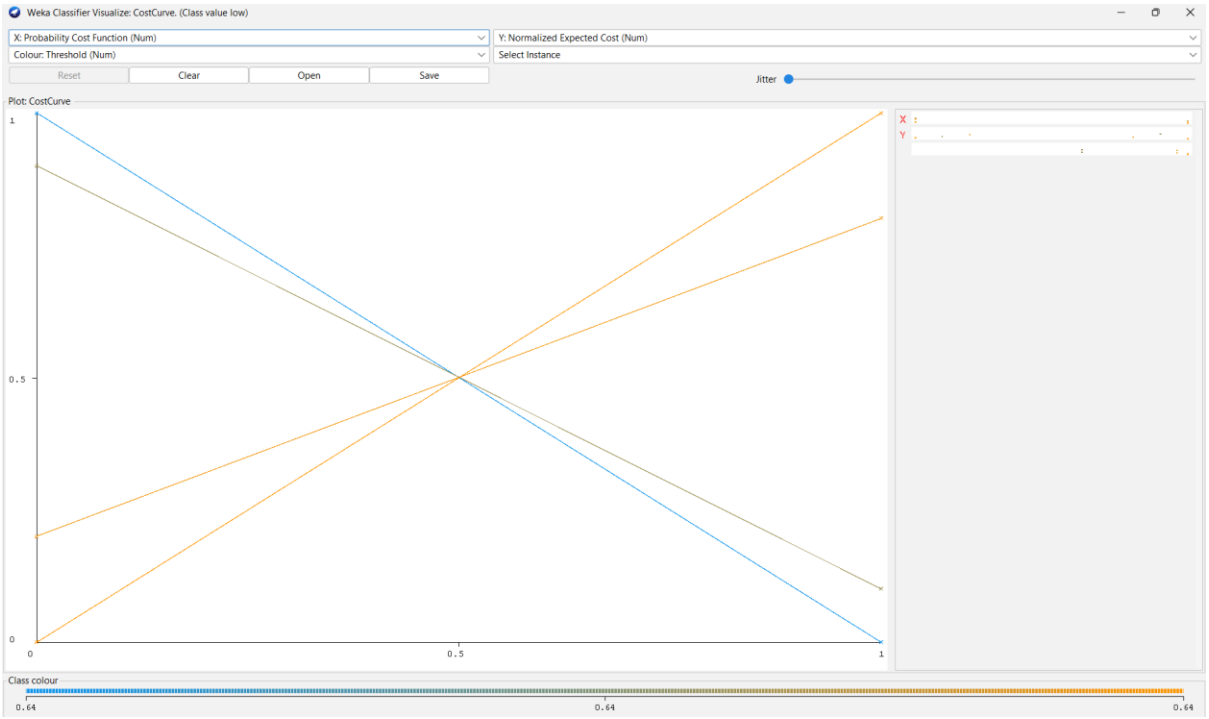

Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Associator
Choose
Apriori -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1 -S -1.0 -c -1
Start
Stop
Result list (right-click for op...
09:13:57 - FilteredAssociator
09:16:19 - Apriori
Associator output
Scheme: weka.associations.Apriori -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1 -S -1.0 -c -1
Relation: supermarket
Instances: 4627
Attributes: 217
[list of attributes omitted]
=== Associator model (full training set) ===
Apriori
=======
Minimum support: 0.15 (694 instances)
Minimum metric <confidence>: 0.9
Number of cycles performed: 17
Generated sets of large itemsets:
Size of set of large itemsets L(1): 44
Size of set of large itemsets L(2): 380
Size of set of large itemsets L(3): 910
Size of set of large itemsets L(4): 633
Size of set of large itemsets L(5): 105
Size of set of large itemsets L(6): 1
Best rules found:
1. biscuits=t frozen foods=t fruit=t total=high 788 ==> bread and cake=t 723 <conf:(0.92)> lift:(1.27) lev:(0.03) [155] conv:(3.35)
2. baking needs=t biscuits=t fruit=t total=high 760 ==> bread and cake=t 696 <conf:(0.92)> lift:(1.27) lev:(0.03) [149] conv:(3.28)
3. baking needs=t frozen foods=t fruit=t total=high 770 ==> bread and cake=t 705 <conf:(0.92)> lift:(1.27) lev:(0.03) [150] conv:(3.27)
4. biscuits=t fruit=t vegetables=t total=high 815 ==> bread and cake=t 746 <conf:(0.92)> lift:(1.27) lev:(0.03) [159] conv:(3.26)
5. party snack foods=t fruit=t total=high 854 ==> bread and cake=t 779 <conf:(0.91)> lift:(1.27) lev:(0.04) [164] conv:(3.15)
6. biscuits=t frozen foods=t vegetables=t total=high 797 ==> bread and cake=t 725 <conf:(0.91)> lift:(1.26) lev:(0.03) [151] conv:(3.06)
7. baking needs=t biscuits=t vegetables=t total=high 772 ==> bread and cake=t 701 <conf:(0.91)> lift:(1.26) lev:(0.03) [145] conv:(3.01)
8. biscuits=t fruit=t total=high 954 ==> bread and cake=t 866 <conf:(0.91)> lift:(1.26) lev:(0.04) [179] conv:(3)
9. frozen foods=t fruit=t vegetables=t total=high 834 ==> bread and cake=t 757 <conf:(0.91)> lift:(1.26) lev:(0.03) [156] conv:(3)
10. frozen foods=t fruit=t total=high 969 ==> bread and cake=t 877 <conf:(0.91)> lift:(1.26) lev:(0.04) [179] conv:(2.92)

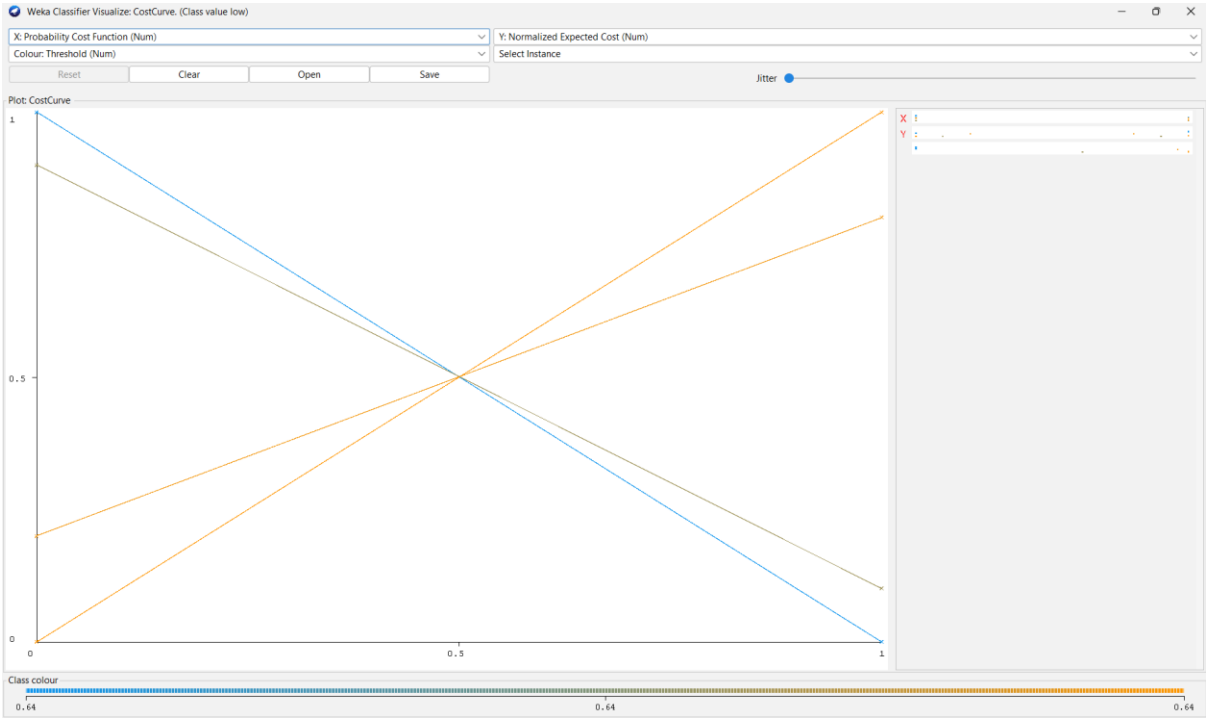
26.

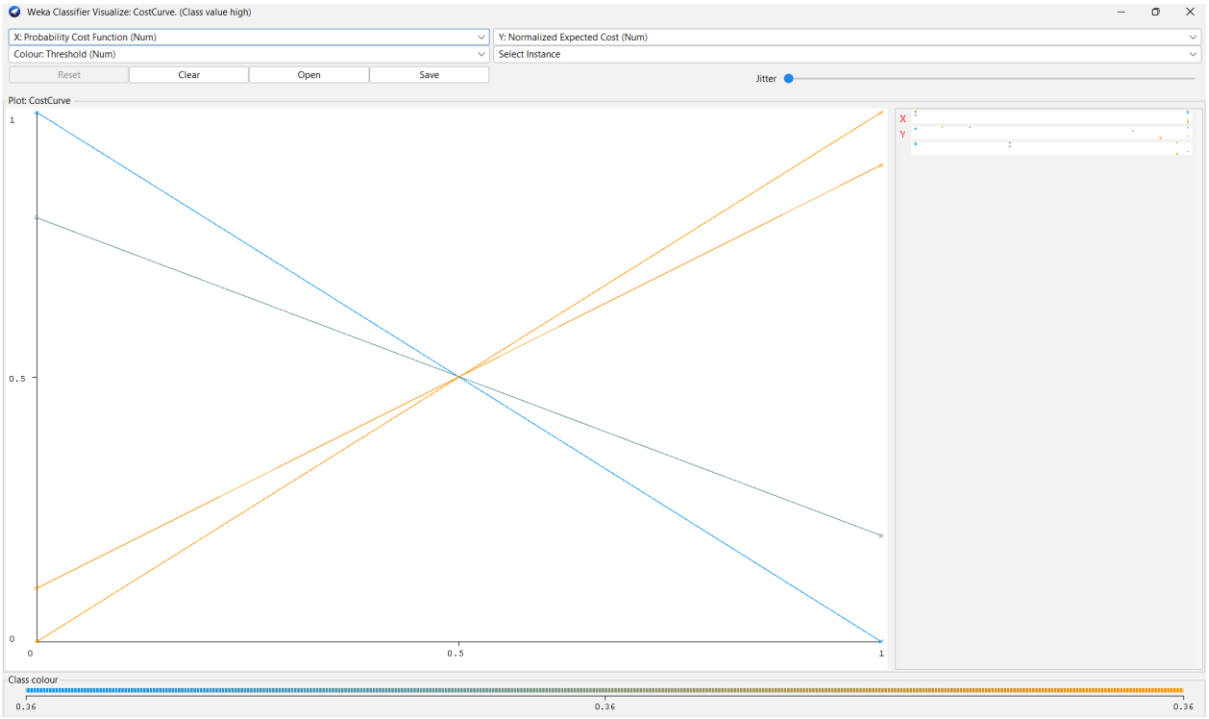

Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Associator
Choose
FPGrowth -P 2 -I -1 -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1
Start
Stop
Result list (right-click for op...
09:13:57 - FilteredAssociator
09:16:19 - Apriori
09:23:15 - FPGrowth
09:23:44 - FPGrowth
Associator output
=== Run information ===
Scheme: weka.associations.FPGrowth -P 2 -I -1 -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1
Relation: supermarket
Instances: 4627
Attributes: 217
[list of attributes omitted]
=== Associator model (full training set) ===
FPGrowth found 16 rules (displaying top 10)
1. [fruit=t, frozen foods=t, biscuits=t, total=high]: 788 ==> [bread and cake=t]: 723 <conf:(0.92)> lift:(1.27) lev:(0.03) conv:(3.35)
2. [fruit=t, baking needs=t, biscuits=t, total=high]: 760 ==> [bread and cake=t]: 696 <conf:(0.92)> lift:(1.27) lev:(0.03) conv:(3.28)
3. [fruit=t, baking needs=t, frozen foods=t, total=high]: 770 ==> [bread and cake=t]: 705 <conf:(0.92)> lift:(1.27) lev:(0.03) conv:(3.27)
4. [fruit=t, vegetables=t, biscuits=t, total=high]: 815 ==> [bread and cake=t]: 746 <conf:(0.92)> lift:(1.27) lev:(0.03) conv:(3.26)
5. [fruit=t, party snack foods=t, total=high]: 854 ==> [bread and cake=t]: 779 <conf:(0.91)> lift:(1.27) lev:(0.04) conv:(3.15)
6. [vegetables=t, frozen foods=t, biscuits=t, total=high]: 797 ==> [bread and cake=t]: 725 <conf:(0.91)> lift:(1.26) lev:(0.03) conv:(3.06)
7. [vegetables=t, baking needs=t, biscuits=t, total=high]: 772 ==> [bread and cake=t]: 701 <conf:(0.91)> lift:(1.26) lev:(0.03) conv:(3.01)
8. [fruit=t, biscuits=t, total=high]: 954 ==> [bread and cake=t]: 866 <conf:(0.91)> lift:(1.26) lev:(0.04) conv:(3)
9. [fruit=t, vegetables=t, frozen foods=t, total=high]: 834 ==> [bread and cake=t]: 757 <conf:(0.91)> lift:(1.26) lev:(0.03) conv:(3)
10. [fruit=t, frozen foods=t, total=high]: 969 ==> [bread and cake=t]: 877 <conf:(0.91)> lift:(1.26) lev:(0.04) conv:(2.92)

27.


Weka Classifier: Cost/Benefit Analysis - trees.J48 (class = high)
X: Sample Size (Num)
Y: True Positive Rate (Num)
Colour: Threshold (Num)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: ThresholdCurve
X: Sample Size (Num)
Y: Cost/Benefit (Num)
Plot: Cost/Benefit Curve
2948
1679
Threshold
% of Population
% of Target (recall)
Score Threshold
% of Population: 0
% of Target: 0
Score Threshold: 0.363
Confusion Matrix
Predicted (a)
Predicted (b)
1679
36.29%
Actual (a): high
2948
63.71%
Actual (b): low
Classification Accuracy: 63.713%
Cost Matrix
Actual (a)
Actual (b)
Total Population: 4627
Cost: 1679
Random: 1679
Gain: 0
Maximize Cost/Benefit
Minimize Cost/Benefit
Cost
Benefit


Weka Classifier Visualize: CostCurve. (Class value low)
X: Probability Cost Function (Num)
Y: Normalized Expected Cost (Num)
Colour: Threshold (Num)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: CostCurve
1
0.5
0
0
0.5
1
X :
Y :
Class colour
0.64
0.64
0.64

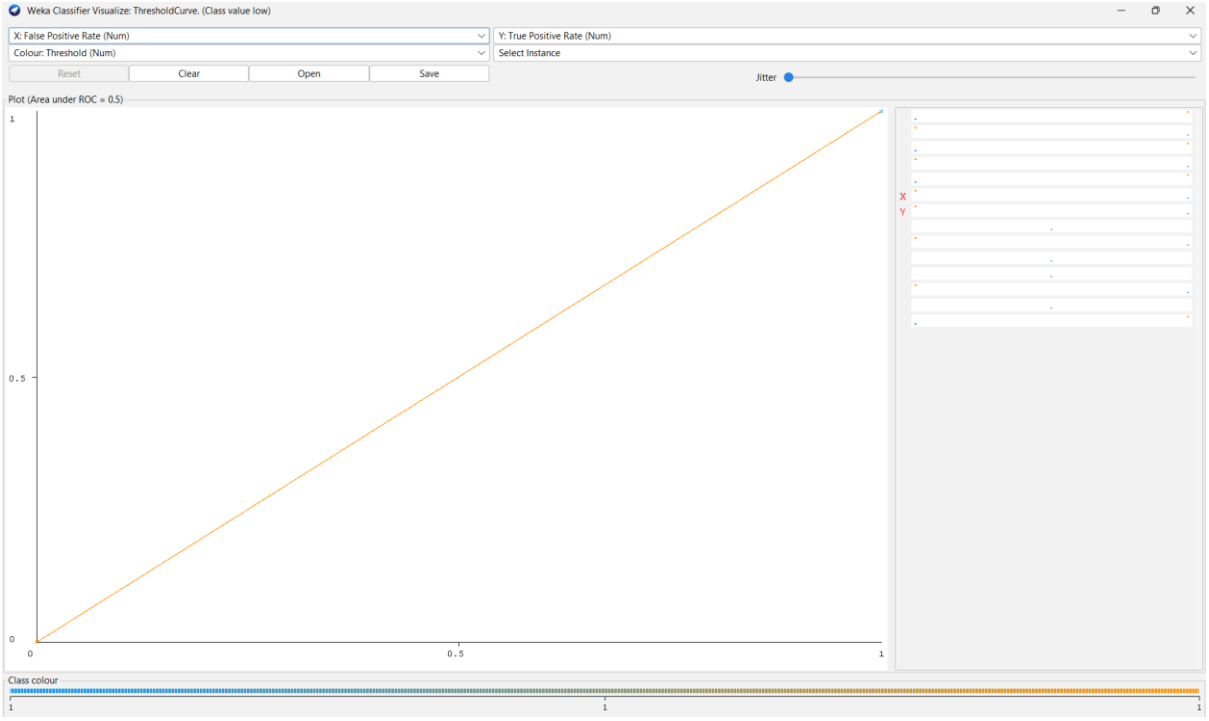

Weka Classifier Visualize: CostCurve. (Class value low)
X: Probability Cost Function (Num)
Y: Normalized Expected Cost (Num)
Colour: Threshold (Num)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: CostCurve
1
0.5
0
0
0.5
1
X :
Y :
Class colour
0.64
0.64
0.64

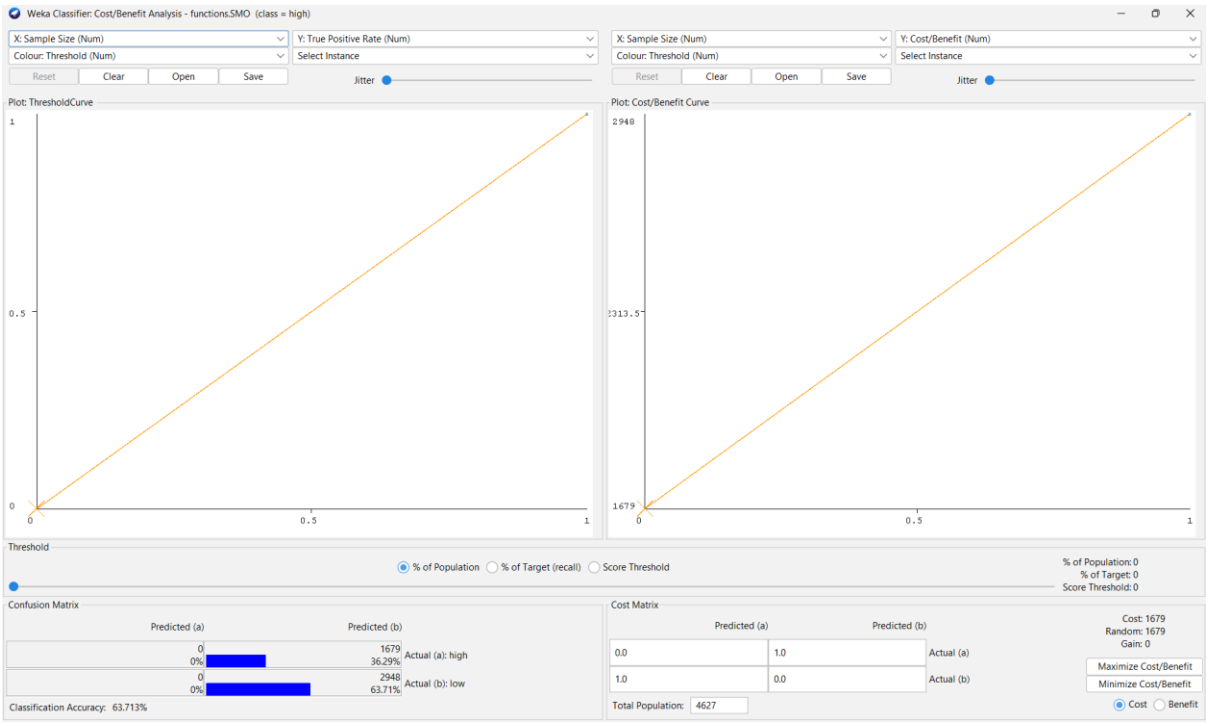

Weka Classifier Visualize: CostCurve. (Class value high)
X: Probability Cost Function (Num)
Y: Normalized Expected Cost (Num)
Colour: Threshold (Num)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: CostCurve
1
0.5
0
0
0.5
1
Class colour
0.36
0.36
0.36

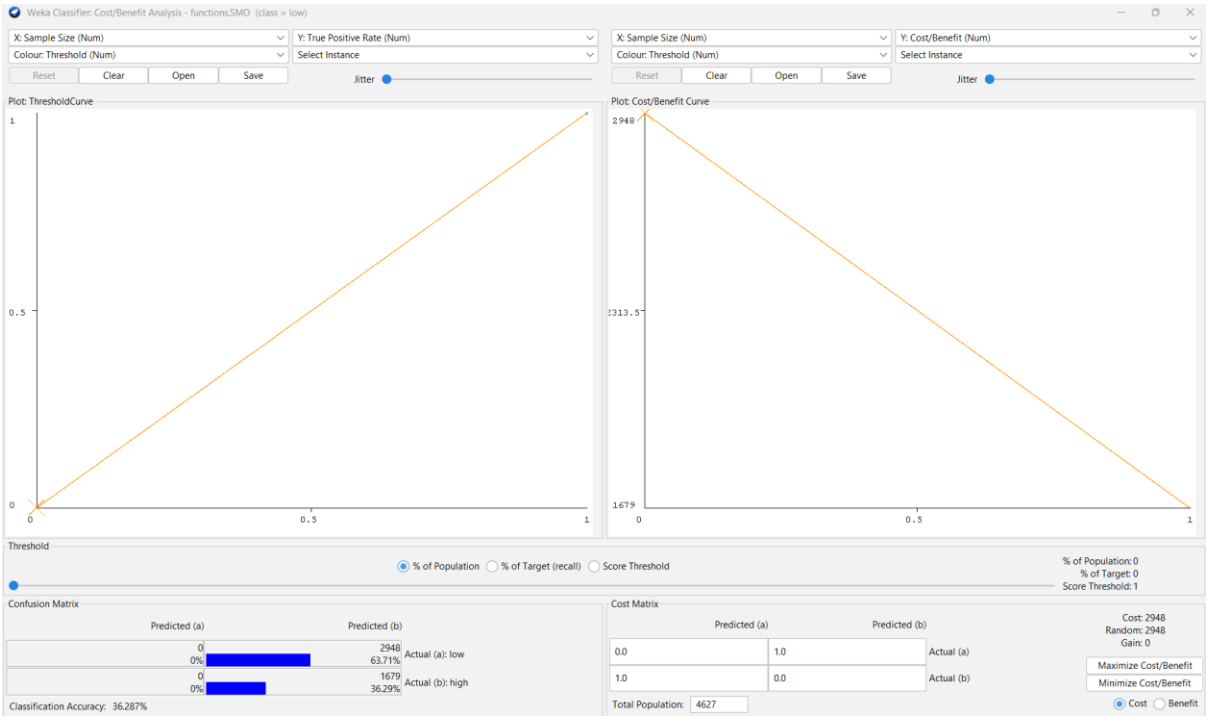
29.


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
Logistic -R 1.0E-8 -M -1 -num-decimal-places 4
Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
10
Percentage split
%
66
More options...
(Nom) total
Start
Stop
Result list (right-click for options)
09:31:12 - trees.J48
09:32:01 - trees.J48
09:46:18 - functions.SMO
09:47:51 - functions.Logistic
Classifier output
Class
Variable low
Intercept 0.5629
Odds Ratios...
Class
Variable low
Time taken to build model: 0.14 seconds
=== Evaluation on training set ===
Time taken to test model on training data: 0.08 seconds
=== Summary ===
Correctly Classified Instances 2948 63.713 %
Incorrectly Classified Instances 1679 36.287 %
Kappa statistic 0
Mean absolute error 0.4624
Root mean squared error 0.4808
Relative absolute error 99.9965 %
Root relative squared error 100 %
Total Number of Instances 4627
=== Detailed Accuracy By Class ===
TP Rate FP Rate Precision Recall F-Measure MCC ROC Area PRC Area Class
1.000 1.000 0.637 1.000 0.778 ? 0.500 0.637 low
0.000 0.000 ? 0.000 ? ? 0.500 0.363 high
Weighted Avg. 0.637 0.637 ? 0.637 ? ? 0.500 0.538
=== Confusion Matrix ===
a b <-- classified as
2948 0 | a = low
1679 0 | b = high
Status
OK
Log
x 0

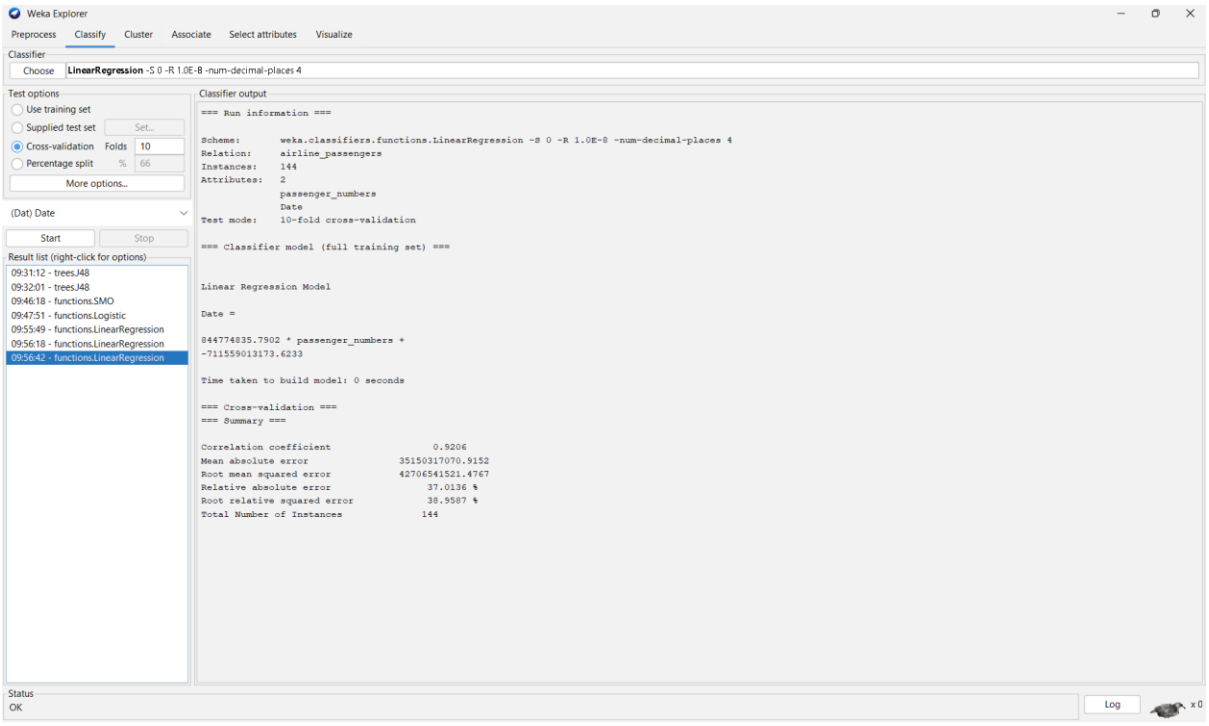
28.

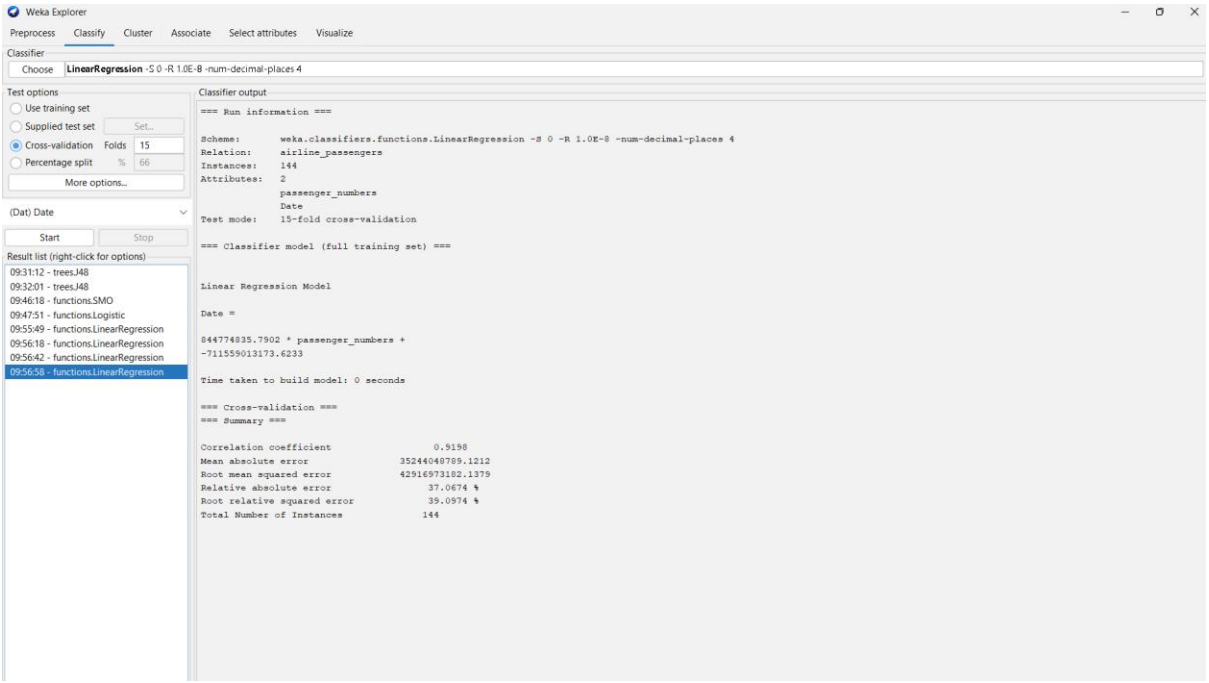

Weka Classifier Visualize: ThresholdCurve. (Class value low)
X: False Positive Rate (Num)
Y: True Positive Rate (Num)
Colour: Threshold (Num)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot (Area under ROC = 0.5)
Class colour

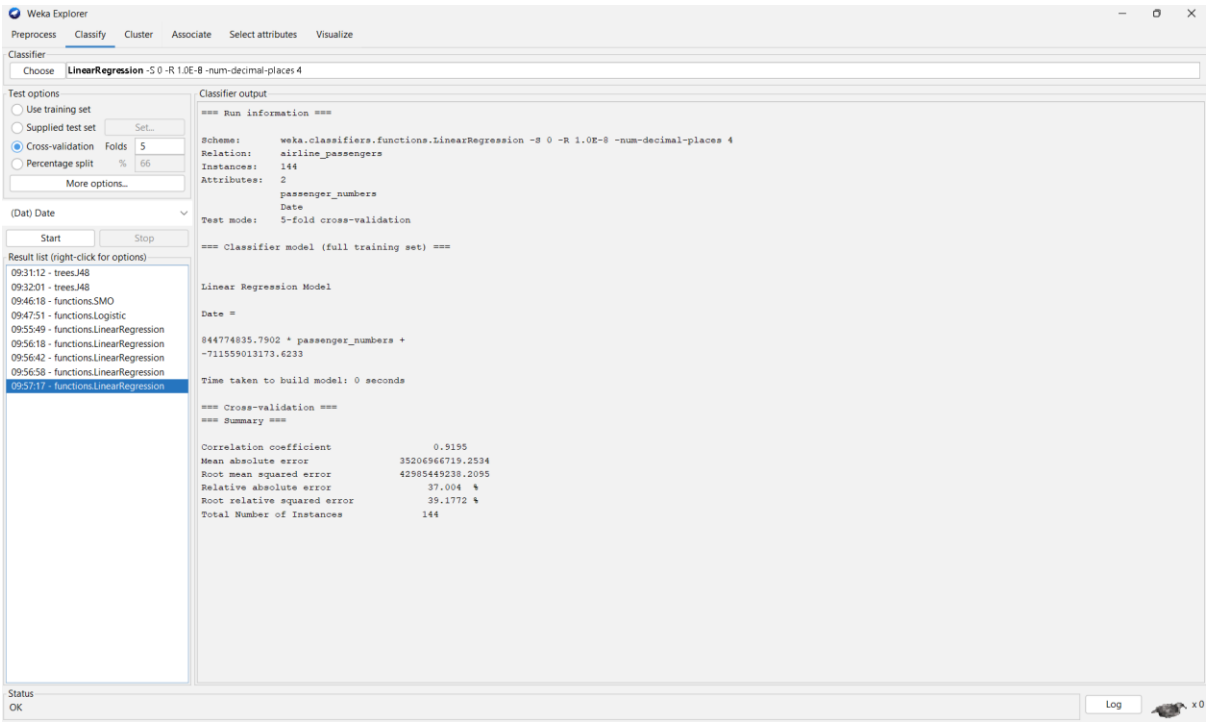

Weka Classifier: Cost/Benefit Analysis - functions.SMO (class = high)
X: Sample Size (Num)
Y: True Positive Rate (Num)
Y: Cost/Benefit (Num)
Colour: Threshold (Num)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: ThresholdCurve
Plot: Cost/Benefit Curve
Threshold
% of Population
% of Target (recall)
Score Threshold
Confusion Matrix
Predicted (a)
Predicted (b)
Actual (a): high
Actual (b): low
Classification Accuracy: 63.713%
Cost Matrix
Actual (a)
Actual (b)
Total Population:
Maximize Cost/Benefit
Minimize Cost/Benefit
Cost
Benefit


Weka Classifier: Cost/Benefit Analysis - functions.SMO (class = low)
X: Sample Size (Num)
Y: True Positive Rate (Num)
Colour: Threshold (Num)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: ThresholdCurve
Y: Cost/Benefit (Num)
Plot: Cost/Benefit Curve
Threshold
% of Population
% of Target (recall)
Score Threshold
% of Population: 0
% of Target: 0
Score Threshold: 1
Confusion Matrix
Predicted (a)
Predicted (b)
Actual (a): low
Actual (b): high
Classification Accuracy: 36.287%
Cost Matrix
Actual (a)
Actual (b)
Total Population: 4627
Cost: 2948
Random: 2948
Gain: 0
Maximize Cost/Benefit
Minimize Cost/Benefit
Cost
Benefit

30.


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
LinearRegression -S 0 -R 1.0E-8 -num-decimal-places 4
Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
10
Percentage split
%
66
More options...
(Dat) Date
Start
Stop
Result list (right-click for options)
09:31:12 - trees.J48
09:32:01 - trees.J48
09:46:18 - functions.SMO
09:47:51 - functions.Logistic
09:55:49 - functions.LinearRegression
09:56:18 - functions.LinearRegression
09:56:42 - functions.LinearRegression
Classifier output
=== Run information ===
Scheme: weka.classifiers.functions.LinearRegression -S 0 -R 1.0E-8 -num-decimal-places 4
Relation: airline_passengers
Instances: 144
Attributes: 2
passenger_numbers
Date
Test mode: 10-fold cross-validation
=== Classifier model (full training set) ===
Linear Regression Model
Date =
844774835.7902 * passenger_numbers +
-711559013173.6233
Time taken to build model: 0 seconds
=== Cross-validation ===
=== Summary ===
Correlation coefficient 0.9206
Mean absolute error 35150317070.9152
Root mean squared error 42706541521.4767
Relative absolute error 37.0136 %
Root relative squared error 38.9587 %
Total Number of Instances 144
Status
OK
Log
x 0


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
LinearRegression -S 0 -R 1.0E-8 -num-decimal-places 4
Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
15
Percentage split
%
66
More options...
(Dat) Date
Start
Stop
Result list (right-click for options)
09:31:12 - trees.J48
09:32:01 - trees.J48
09:46:18 - functions.SMO
09:47:51 - functions.Logistic
09:55:49 - functions.LinearRegression
09:56:18 - functions.LinearRegression
09:56:42 - functions.LinearRegression
09:56:58 - functions.LinearRegression
Classifier output
=== Run information ===
Scheme: weka.classifiers.functions.LinearRegression -S 0 -R 1.0E-8 -num-decimal-places 4
Relation: airline_passengers
Instances: 144
Attributes: 2
passenger_numbers
Date
Test mode: 15-fold cross-validation
=== Classifier model (full training set) ===
Linear Regression Model
Date =
844774835.7902 * passenger_numbers +
-711559013173.6233
Time taken to build model: 0 seconds
=== Cross-validation ===
=== Summary ===
Correlation coefficient 0.9198
Mean absolute error 35244048789.1212
Root mean squared error 42916973182.1379
Relative absolute error 37.0674 %
Root relative squared error 39.0974 %
Total Number of Instances 144


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Classifier
Choose
LinearRegression -S 0 -R 1.0E-8 -num-decimal-places 4
Test options
Use training set
Supplied test set
Set...
Cross-validation
Folds
5
Percentage split
%
66
More options...
(Dat) Date
Start
Stop
Result list (right-click for options)
09:31:12 - trees.J48
09:32:01 - trees.J48
09:46:18 - functions.SMO
09:47:51 - functions.Logistic
09:55:49 - functions.LinearRegression
09:56:18 - functions.LinearRegression
09:56:42 - functions.LinearRegression
09:56:58 - functions.LinearRegression
09:57:17 - functions.LinearRegression
Classifier output
=== Run information ===
Scheme: weka.classifiers.functions.LinearRegression -S 0 -R 1.0E-8 -num-decimal-places 4
Relation: airline_passengers
Instances: 144
Attributes: 2
passenger_numbers
Date
Test mode: 5-fold cross-validation
=== Classifier model (full training set) ===
Linear Regression Model
Date =
844774835.7902 * passenger_numbers +
-711559013173.6233
Time taken to build model: 0 seconds
=== Cross-validation ===
=== Summary ===
Correlation coefficient 0.9195
Mean absolute error 35206966719.2534
Root mean squared error 42985449238.2055
Relative absolute error 37.004 %
Root relative squared error 39.1772 %
Total Number of Instances 144
Status
OK
Log
x 0